Accuracy of malaria diagnostic tests performed on non-invasively collected samples: a systematic review and meta-analysis

APPENDIX

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Supplementary Table 1. Search strategy for PubMed

Search	**Search terms**
#1	"non-invasive"[tiab] OR "non-blood"[tiab] OR "urine"[tiab] OR "saliva"[tiab]
#2	"malaria"[MeSH Terms]
#3	#1 AND #2
Date: 19 January 2021, 422 citations Restrictions: No	

Supplementary Table 2. Search strategy for EMBASE

Search	**Search terms**
#1	'non invasive':ti,ab,kw OR 'non blood':ti,ab,kw OR urine:ti,ab,kw OR saliva:ti,ab,kw
#2	malaria:ti,ab,kw
#3	#1 AND #2
Date: 19 January 2021, 1185 citations Restrictions: No	

Supplementary Table 3. General characteristics of included studies

Characteristics		N	%
Country where the study was conducted	Iran	10	27.8%
	Nigeria	5	13.9%
	India	5	13.9%
	Cameroon	4	11.1%
	The Gambia	4	11.1%
	Ghana	3	8.3%
	Thailand	2	5.6%
	Rwanda	1	2.8%
	Uganda	1	2.8%
	Zambia	1	2.8%
Settings where the study was conducted	Hospital	31	86.1%
	Community	3	8.3%
	Unclear	2	5.5%
Region where the study was conducted	Africa	20	55.5%
	Outside Africa	16	44.4%
Study design	Cross sectional	33	91.7%
	Case control	3	8.3%
Age groups	Children	5	13.9%
	All age categories	28	77.7%
	Unclear	3	8.3%
Period during which studies were conducted		2008-2020	
Age range (yrs, n=24 studies)		1-80	
Male % (n=27 studies)		29-82.4	

Supplementary Table 4. Individual characteristics of the included studies

Author	Year	Design	Country	Setting	Mean Age	Minimum age	Maximum age	Male %	Characteristic of the population in terms of age and diseases/conditions	Characteristics of patients in terms of fever or other malaria symptoms	Name of the reference (GOLD STANDARD) test	Name of the material for non-invasive test	Name of the test performed on the non-invasive sample	Details about the test performed on the non-invasive sample	TP	FP	TN	FN	Person(s) performing de analysis	Plasmodium Target by the non-invasive test	Transport and storage conditions for the non-invasive sample	Parasite density
Singh_1	2014	Cross sectional	India	Hospital based	NR	1	80	67	All age categories and conditions*	Symptomatic	Microscopy	Saliva	PCR	Nested PCR	83	2	121	12	Experienced worker	Both falciparum and non-falciparum	A sample of 0.5 to 2.0 ml of saliva, collected in sterile plastic vials by the passive drool method immediately after blood collection and stored at 4 °C	Overall parasite density for P. vivax : from 320 to 61,600/µl (geometric mean 4,941 parasites/µl) and P. falciparum 200 to 496,000 (geometric mean 21,850 parasites/µl).
Singh_2	2014	Cross sectional	India	Hospital based	NR	1	80	67	All age categories and conditions	Symptomatic	Microscopy	Saliva	PCR	Singleplex CRS PCR	77	2	121	18	Experienced worker	Both falciparum and non-falciparum	A sample of 0.5 to 2.0 ml of saliva, collected in sterile plastic vials by the passive drool method immediately after blood collection and stored at 4 °C	Overall parasite density for P. vivax : from 320 to 61,600/µl (geometric mean 4,941 parasites/µl) and P. falciparum 200 to 496,000 (geometric mean 21,850 parasites/µl).
Singh_3	2014	Cross sectional	India	Hospital based	NR	1	80	67	All age categories and conditions	Symptomatic	Microscopy	Saliva	PCR	Multiplex CRS PCR	67	1	122	28	Experienced worker	Both falciparum and non-falciparum	A sample of 0.5 to 2.0 ml of saliva, collected in sterile plastic vials by the passive drool method immediately after blood collection and stored at 4 °C	Overall parasite density for P. vivax : from 320 to 61,600/µl (geometric mean 4,941 parasites/µl) and P. falciparum 200 to 496,000 (geometric mean 21,850 parasites/µl).
Ghayour_1	2014	Cross sectional	Iran	Hospital based	NR	NR	NR	NR	All age categories and conditions	Symptomatic	PCR	Urine	PCR		46	1	32	20	Unclear	Both falciparum and non-falciparum	2 ml of blood and 5 ml of urine stored in ethanol at -20°C.	Unclear
Ghayour_2	2014	Cross sectional	Iran	Hospital based	NR	NR	NR	NR	All age categories and conditions	Symptomatic	PCR	Saliva	PCR		60	1	32	6	Unclear	Both falciparum and non-falciparum	2 ml of blood and 2ml of saliva stored in ethanol at -20°C.	Unclear
Mfuh_1	2017	Cross sectional	Cameroon	Hospital based	22	2	76	29	All age categories and conditions	Symptomatic	Microscopy	Saliva	PCR		50	12	157	3	Experienced worker	Falciparum	1 ml of saliva into OMNIgene ®•ORAL (OM-501) kit, stored at room temperature until DNA extraction 2–6 weeks later	Median parasitaemia by age group/ µl of blood: 71,907 (2–10 years); 23,257 (11–20); 26,760 (21–39); 13,600 (> 40)
Mfuh_2	2017	Cross sectional	Cameroon	Hospital based	22	2	76	29	All age categories and conditions	Symptomatic	PCR	Saliva	PCR		64	1	143	14	Experienced worker	Falciparum	1 ml of saliva into OMNIgene ®•ORAL (OM-501) kit, stored at room temperature until DNA extraction 2–6 weeks later	Median parasitaemia by age group/ µl of blood: 71,907 (2–10 years); 23,257 (11–20); 26,760 (21–39); 13,600 (> 40)
Nwakanma_1	2009	Cross sectional	The Gambia	Hospital based	25	10	45	63.2	All age categories and conditions	Symptomatic	Microscopy	Saliva	PCR		37	10	325	14	Experienced worker	Both falciparum and non-falciparum	2-mL of saliva collected into sterile bottles and stored at 4°C–8°C until DNA, usually within 2 h	Unclear
Nwakanma_2	2009	Cross sectional	The Gambia	Hospital based	25	10	45	63.2	All age categories and conditions	Symptomatic	Microscopy	Urine	PCR		16	6	328	34	Experienced worker	Both falciparum and non-falciparum	5-mL of urine collected into sterile bottles and stored at 4°C–8°C until DNA, usually within 2 h.	Unclear

Wilson	2008	Cross sectional	Ghana	Hospital based	NR	1.8	16	42.5	Children	Symptomatic	Microscopy	Saliva	ELISA		13	0	10	17	Unclear	Falciparum	Saliva collected in sterile containers and aliquoted into microcentrifuge tubes and stored at −20°C	Unclear
Aninagyei_1	2020	Cross sectional	Ghana	Hospital based	22.3	6	39	42.7	All age categories and conditions	Symptomatic	RDT	Urine	RDT		22	0	325	269	Unclear	Both falciparum and non-falciparum	30 mL of urine frozen at − 20 °C	Unclear
Aninagyei_2	2020	Cross sectional	Ghana	Hospital based	22.3	6	39	42.7	All age categories and conditions	Symptomatic	RDT	Saliva	RDT		28	0	325	182	Unclear	Both falciparum and non-falciparum	5 mL of saliva frozen at − 20 °C	Unclear
Netongo_1	2019	Cross sectional	Cameroon	Hospital based	27	2	77	24.1	All age categories and conditions	Symptomatic	Microscopy	Saliva	PCR	homemade kit (Formulation f1) to stabilize Plasmodium DNA in saliva stored at room temperature	65	3	15	0	Experienced worker	Both falciparum and non-falciparum	1–2 ml of saliva introduced into equal volume of the homemade DNA stabilization buffer	Not reported
Netongo_2	2019	Cross sectional	Cameroon	Hospital based	27	2	77	24.1	All age categories and conditions	Symptomatic	Microscopy	Saliva	PCR	OMNIgene ® ORAL (S0)	65	4	14	0	Experienced worker	Both falciparum and non-falciparum	1–2 ml of saliva introduced into equal volume of the OMNIgene®ORAL (OM-501) kit (S0)	Not reported
AlShehri	2019	Cross sectional	Uganda	Population-based	6.8	6.6	6.9	50.2	Unclear	Asymptomatic	RDT	Stool	PCR		101	57	52	37	Unclear	Both falciparum and non-falciparum	Two stool samples per participant, filtered through a 212 µm metal mesh then 0.5 g placed in 1 mL of 95% ethanol for the EPF before transfer to the to a foreign lab for DNA extraction	Not reported
Samal	2017	Case control study	India	Hospital based	NR	NR	NR	NR	Unclear	Symptomatic	Microscopy	Urine	RDT		52	3	48	8	Experienced worker	Falciparum	1 ml of urine in a sterile tube. Storage conditions unclear	Unclear
Oyibo_1	2014	Cross sectional	Nigeria	Unclear/No description	18.6	2	80	43	All age categories and conditions	Asymptomatic	Microscopy	Urine	RDT		98	79	909	39	Unclear	Falciparum	Unclear	Unclear
Oyibo_2	2014	Cross sectional	Nigeria	Unclear/No description	18.6	2	80	43	All age categories and conditions	Symptomatic	Microscopy	Urine	RDT		173	58	304	31	Unclear	Falciparum	Unclear	Unclear
Najafabadi_1	2014	Cross sectional	Iran	Hospital based	28	3	57	82.4	All age categories and conditions	Symptomatic	Microscopy	Saliva	PCR		63	1	38	6	Experienced worker	Both falciparum and non-falciparum	Two volumes of absolute ethanol added to 2 ml of saliva, transported to the laboratory and stored at −20◦C	Geometric mean parasite density of positive samples = 3970.7 parasites/ µl (range = 120–94,117 parasites/ µl).
Najafabadi_2	2014	Cross sectional	Iran	Hospital based	28	3	57	82.4	All age categories and conditions	Symptomatic	Microscopy	Urine	PCR		50	0	39	19	Experienced worker	Both falciparum and non-falciparum	Two volumes of absolute ethanol added to 5 ml of urine, transported to the laboratory and stored at −20◦C	Geometric mean parasite density of positive samples = 3970.7 parasites/ µl (range = 120–94,117 parasites/ µl).
Najafabadi_3	2014	Cross sectional	Iran	Hospital based	28	3	57	82.4	All age categories and conditions	Symptomatic	Microscopy	Saliva	LAMP		33	0	39	36	Experienced worker	Both falciparum and non-falciparum	Two volumes of absolute ethanol added to 2 ml of saliva, transported to the laboratory and stored at −20◦C	Geometric mean parasite density of positive samples = 3970.7 parasites/ µl (range = 120–94,117 parasites/ µl).
Najafabadi_4	2014	Cross sectional	Iran	Hospital based	28	3	57	82.4	All age categories and conditions	Symptomatic	Microscopy	Urine	LAMP		21	0	39	48	Experienced worker	Both falciparum and non-falciparum	Two volumes of absolute ethanol added to 5 ml of urine, transported to the laboratory and stored at −20◦C	Geometric mean parasite density of positive samples = 3970.7 parasites/ µl (range = 120–94,117 parasites/ µl).

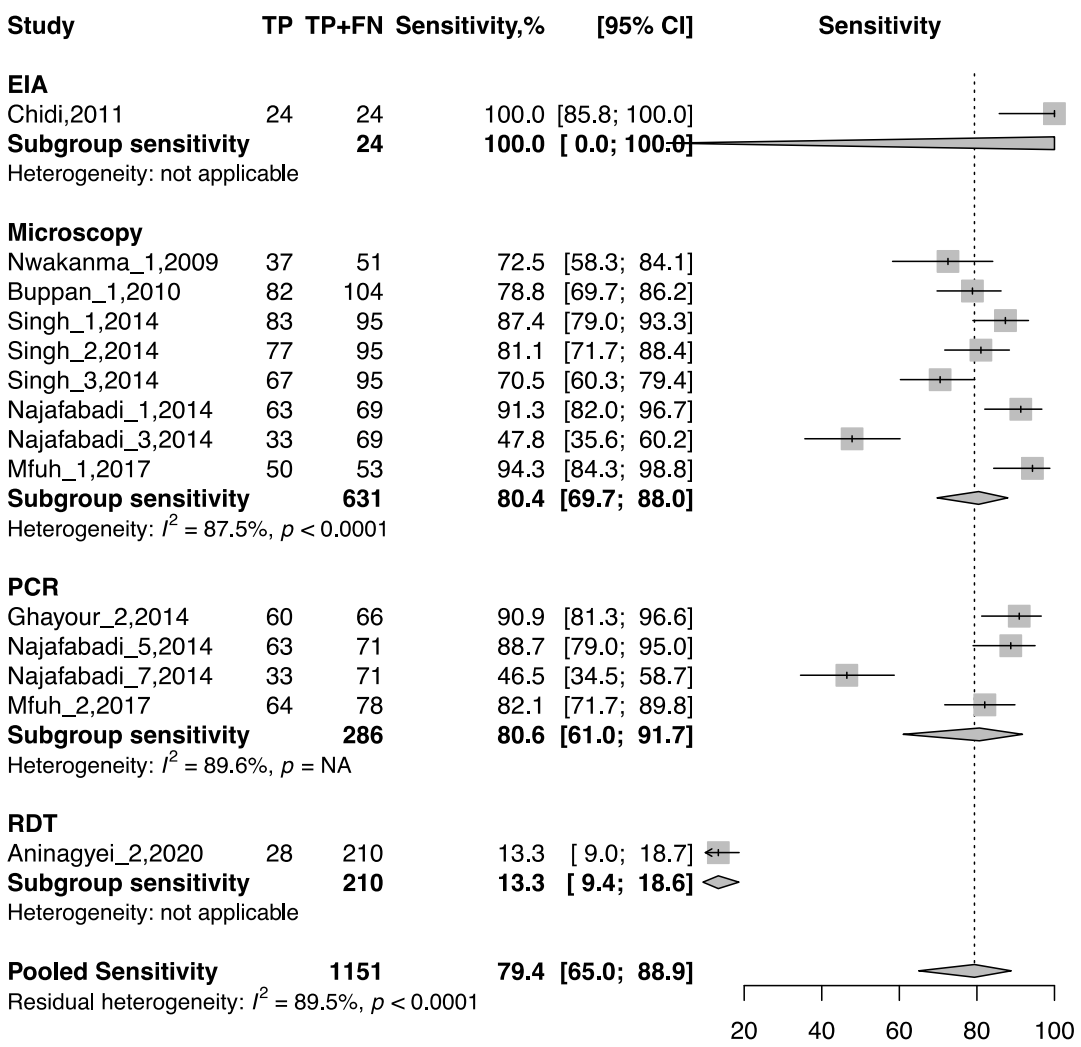
Najafabadi_5	2014	Cross sectional	Iran	Hospital based	28	3	57	82.4	All age categories and conditions	Symptomatic	PCR	Saliva	PCR		63	1	36	8	Experienced worker	Both falciparum and non-falciparum	Two volumes of absolute ethanol added to 2 ml of saliva, transported to the laboratory and stored at −20◦C	Geometric mean parasite density of positive samples = 3970.7 parasites/ µl (range = 120–94,117 parasites/ µl).
Najafabadi_6	2014	Cross sectional	Iran	Hospital based	28	3	57	82.4	All age categories and conditions	Symptomatic	PCR	Urine	PCR		50	0	37	21	Experienced worker	Both falciparum and non-falciparum	Two volumes of absolute ethanol added to 5 ml of urine, transported to the laboratory and stored at −20◦C	Geometric mean parasite density of positive samples = 3970.7 parasites/ µl (range = 120–94,117 parasites/ µl).
Najafabadi_7	2014	Cross sectional	Iran	Hospital based	28	3	57	82.4	All age categories and conditions	Symptomatic	PCR	Saliva	LAMP		33	0	37	38	Experienced worker	Both falciparum and non-falciparum	Two volumes of absolute ethanol added to 2 ml of saliva, transported to the laboratory and stored at −20◦C	Geometric mean parasite density of positive samples = 3970.7 parasites/ µl (range = 120–94,117 parasites/ µl).
Najafabadi_8	2014	Cross sectional	Iran	Hospital based	28	3	57	82.4	All age categories and conditions	Symptomatic	PCR	Urine	LAMP		21	0	37	50	Experienced worker	Both falciparum and non-falciparum	Two volumes of absolute ethanol added to 5 ml of urine, transported to the laboratory and stored at −20◦C	Geometric mean parasite density of positive samples = 3970.7 parasites/ µl (range = 120–94,117 parasites/ µl).
Oguonu	2014	Cross sectional	Nigeria	Hospital based	NR	NR	NR	NR	All age categories and conditions	Symptomatic	Microscopy	Urine	RDT		67	19	96	13	Experienced worker	Falciparum	NR	
Gbotosho_1	2010	Case control study	Nigeria	Hospital based	7.2	0.7	13	NR	Children	Symptomatic	Microscopy	Saliva	PCR		30	2	2	3	Unclear	Falciparum	Whole saliva analysed immediately, or stored at 4°C for 24 hours before analysis	Geometric mean parasite density was 59,179 asexual parasites/µL blood (range =2,463–551,614)
Gbotosho_2	2010	Cross sectional	Nigeria	Hospital based	7.2	0.7	13	NR	Children	Symptomatic	Microscopy	Saliva	RDT		53	0	10	15	Unclear	Falciparum	Whole saliva analysed immediately, or stored at 4°C for 24 hours before analysis	Geometric mean parasite density was 59,179 asexual parasites/µL blood (range = 2,463–551,614)
Buppan_1	2010	Cross sectional	Thailand	Hospital based	20.3	4	60	60.8	All age categories and conditions	Symptomatic	Microscopy	Saliva	PCR		82	11	125	22	Experienced worker	Both falciparum and non-falciparum	1 - 2 ml of saliva, half volume of which was kept on ice during transportation to the lab, and the remaining was preserved by two volumes of absolute ethanol and kept at 25°C - 35°C	Overall parasite density of microscopy positive samples ranged=35-311,395 parasites/ml (geometric mean = 13,920 parasites/ml).
Buppan_2	2010	Cross sectional	Thailand	Hospital based	20.3	4	60	20.3	All age categories and conditions	Symptomatic	Microscopy	Urine	PCR		41	1	135	63	Experienced worker	Both falciparum and non-falciparum	20 ml of midstream urine, half volume of which was kept on ice during transportation to the lab, and the remaining was preserved by two volumes of absolute ethanol and kept at 25°C - 35°C	Overall parasite density of microscopy positive samples ranged=35-311,395 parasites/ml (geometric mean = 13,920 parasites/ml).
Guest_1	2019	Cross sectional	The Gambia	Population-based	NR	5	13	NR	Children	Asymptomatic	Microscopy/PCR	Skin odour	Dog smell	Dog L	22	13	132	8	Dog	Unclear	NA	Unclear
Guest_2	2019	Cross sectional	The Gambia	Population-based	NR	5	13	NR	Children	Asymptomatic	Microscopy/PCR	Skin odour	Dog smell	Dog S	21	14	131	9	Dog	Unclear	NA	Unclear
Olasehinde	2016	Case control study	India	Hospital based	NR	NR	NR	NR	Unclear	Symptomatic	Microscopy	Urine	RDT		13	0	25	62	Two pathologists	Both falciparum and non-falciparum	Unclear	
Chidi	2011	Cross sectional	Zambia	Hospital based	13.5	1	80	55	All age categories and conditions	Unclear	EIA	Saliva	EIA		24	0	29	0	Unclear	Falciparum	Participants' gums were swabbed, and samples were transferred to collection tube with transport buffer	Unclear

																				and stored at room temperature	
Gómez-Luque	2020	Cross sectional	Rwanda	Hospital based	NR	4	31	47.4	All age categories and conditions	Symptomatic	RDT and qPCR	Hair	qPCR		-	-	-	-	Unclear	Falciparum	50 units of hair without root keep in in sterile resealable plastic bags and at room temperature (4 months at 18–24 °C), refrigeration (3 months at 2–6 °C) and freezing (1 month at -20 °C and 1 month at -80 °C)	Unclear

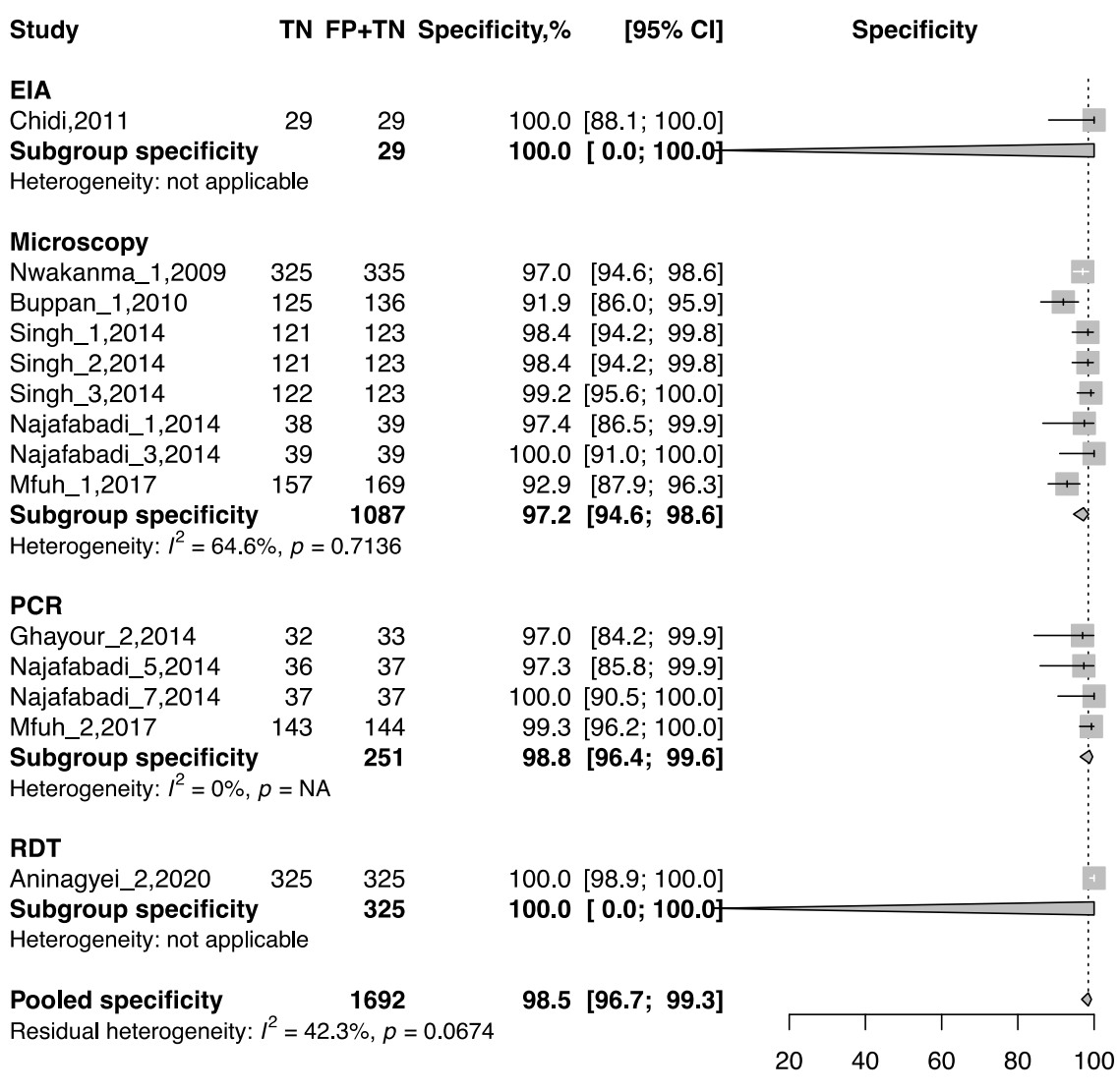
**Condition refers to pregnancy/post-partum or diseases such as sickle cell disease or any other chronic (know or unknow) disease.*

SALIVA

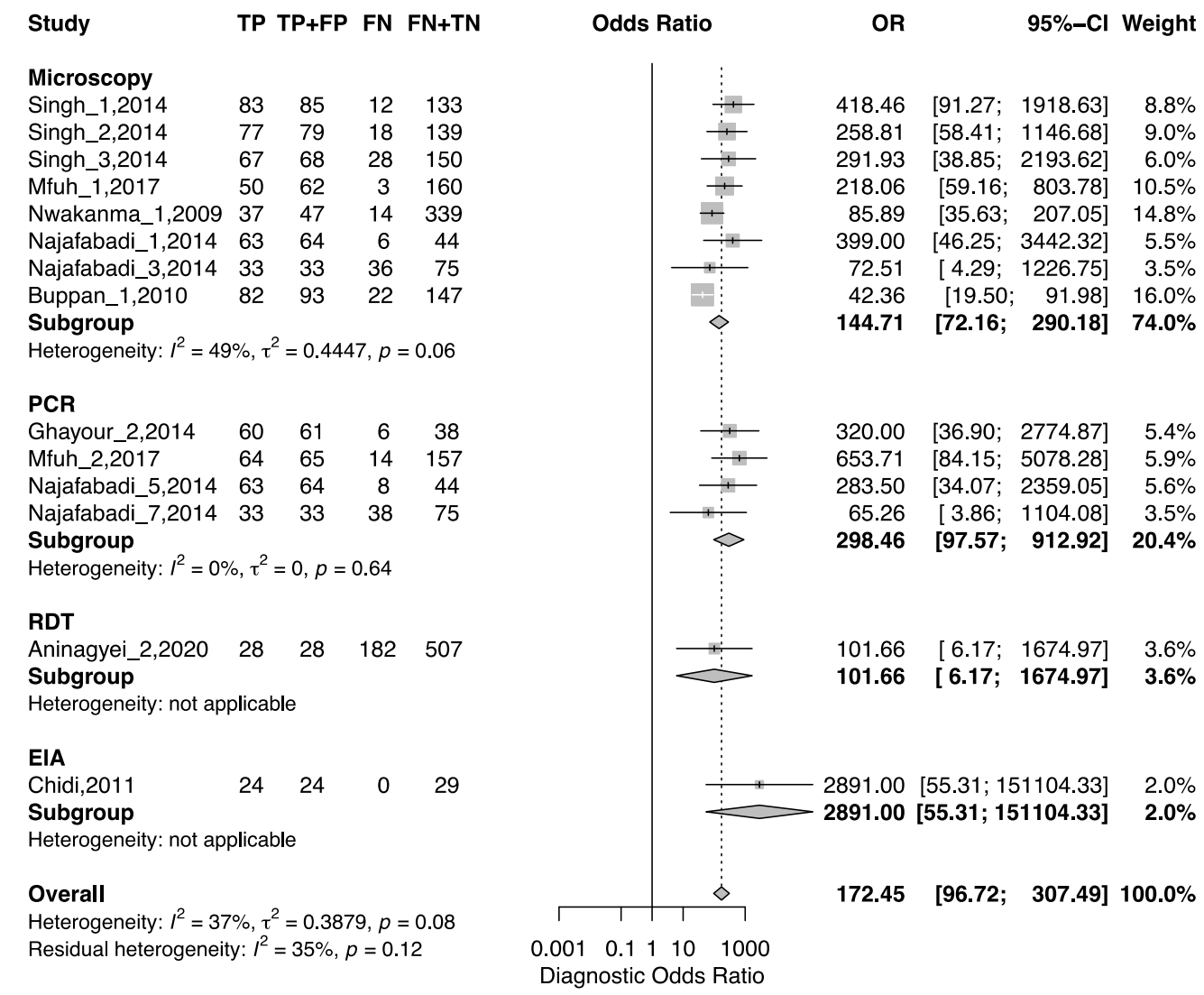
Diagnostic accuracy of tests conducted on saliva according to the reference test performed on blood



Supplementary Figure 1. Sensitivity of tests conducted on saliva according to the reference test performed on blood



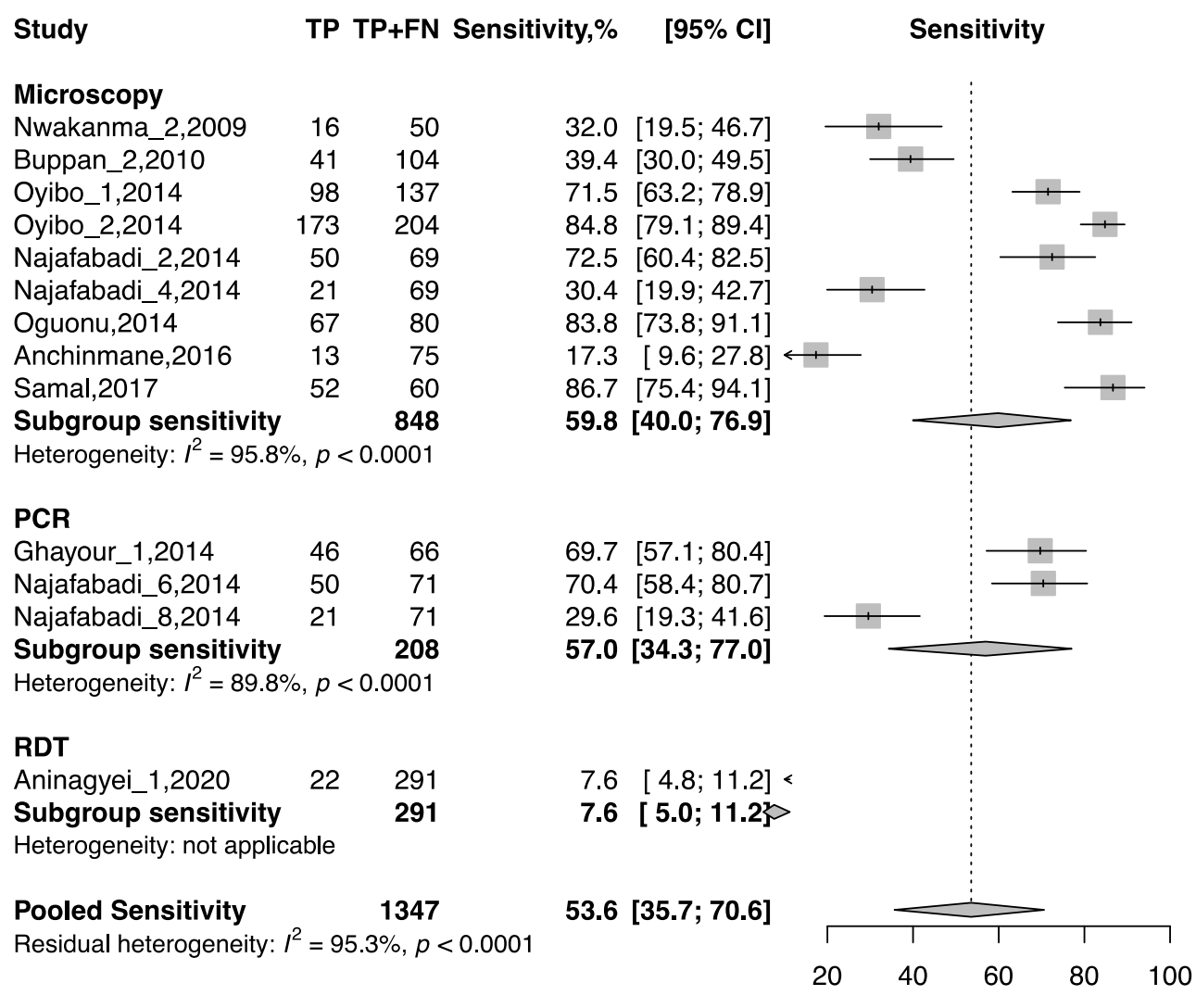
Supplementary Figure 2. Specificity of tests conducted on saliva according to the reference test performed on blood



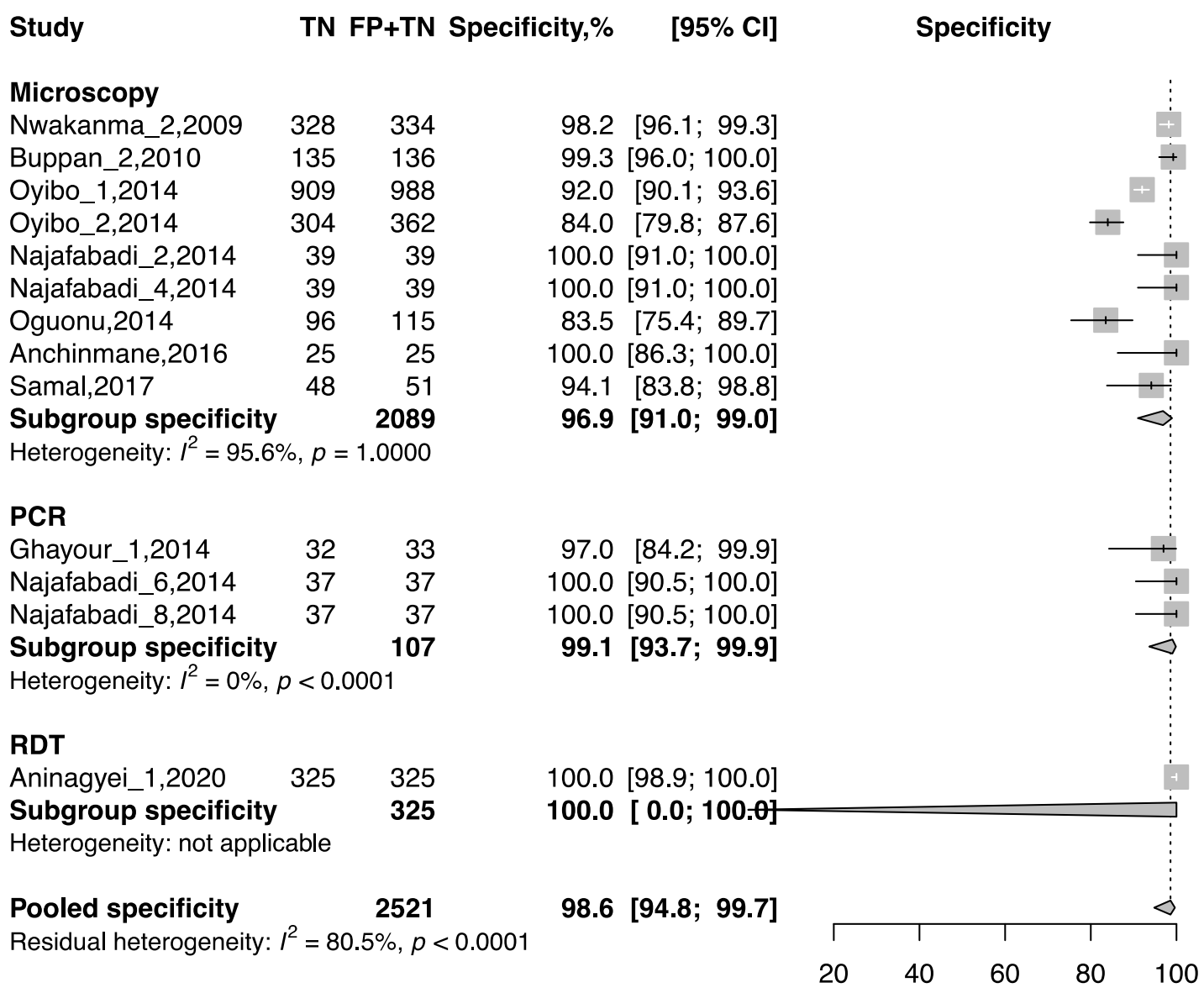
Supplementary Figure 3. DOR of tests conducted on saliva according to the reference test performed on blood

Urine

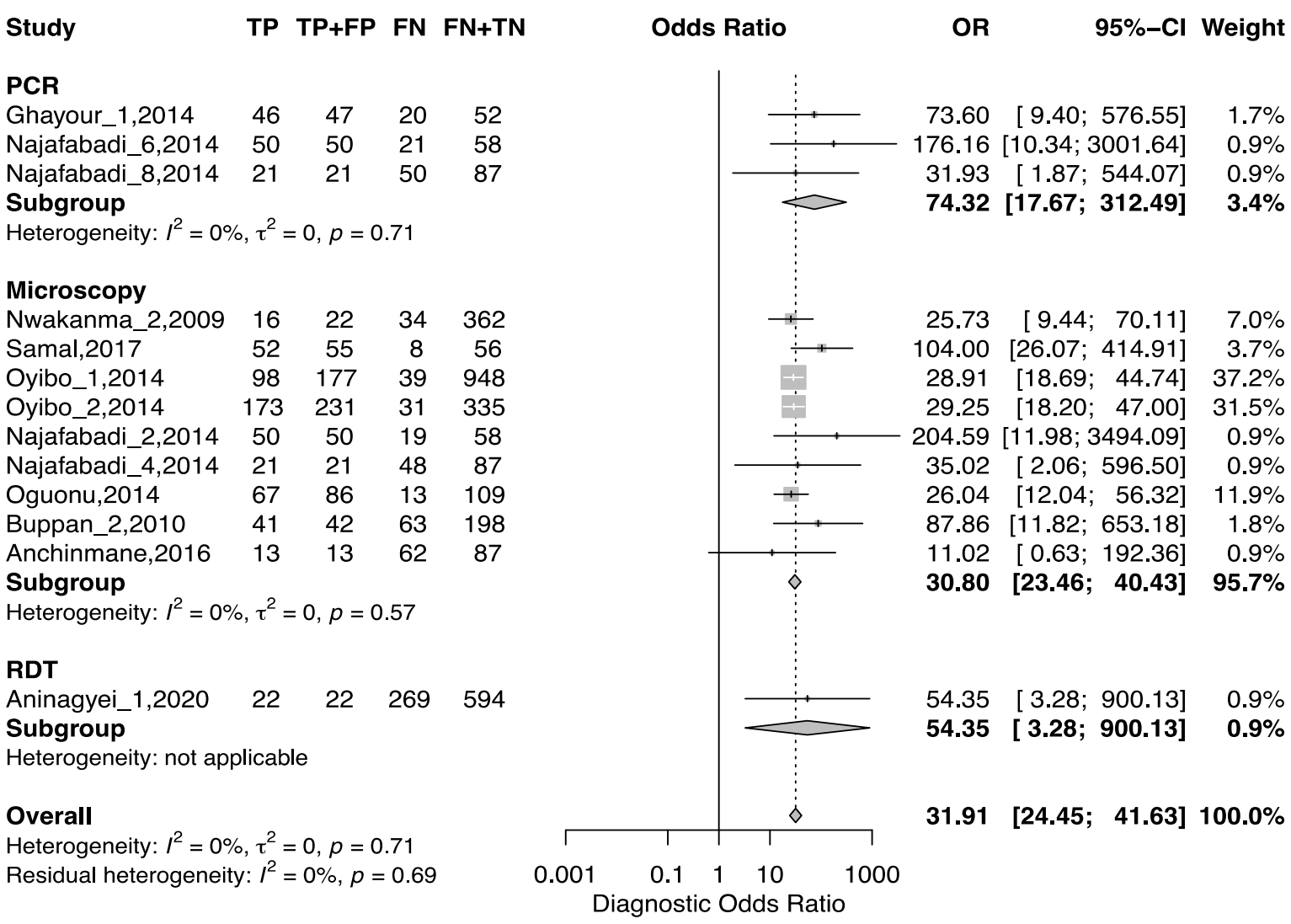
Diagnostic accuracy of tests conducted on urine according to the reference test perform on blood sample



Supplementary Figure 4. Sensitivity of tests conducted on urine according to the reference test perform on blood sample



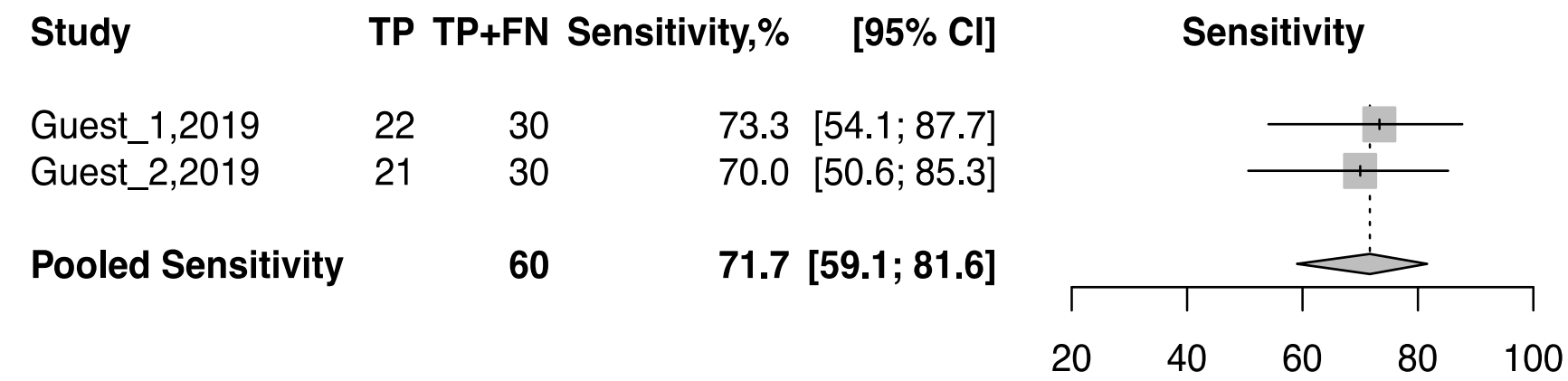
Supplementary Figure 5. Specificity of tests conducted on urine according to the reference test perform on blood sample



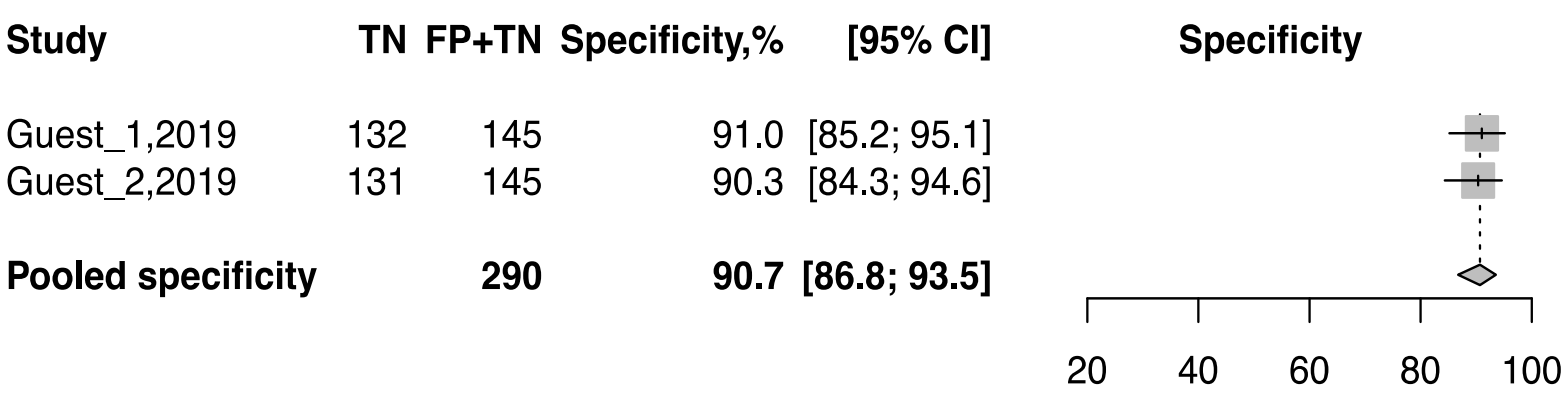
Supplementary Figure 6. DOR of tests conducted on urine according to the reference test perform on blood sample

Studies conducted in children

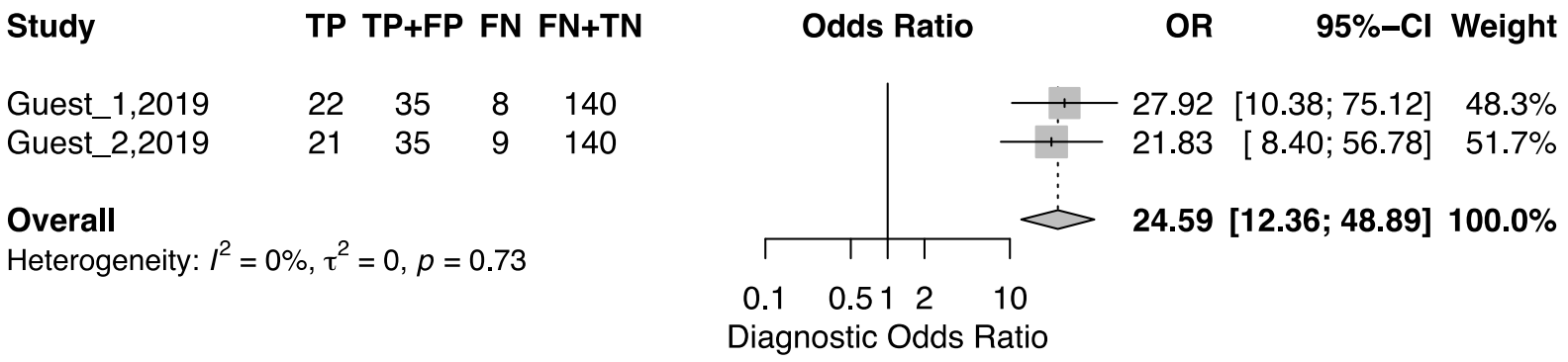
Studies conducted in children: only study on the "sniff and tell" method were eligible for meta-analysis



Supplementary Figure 7. Sensitivity of "sniff and tell" method



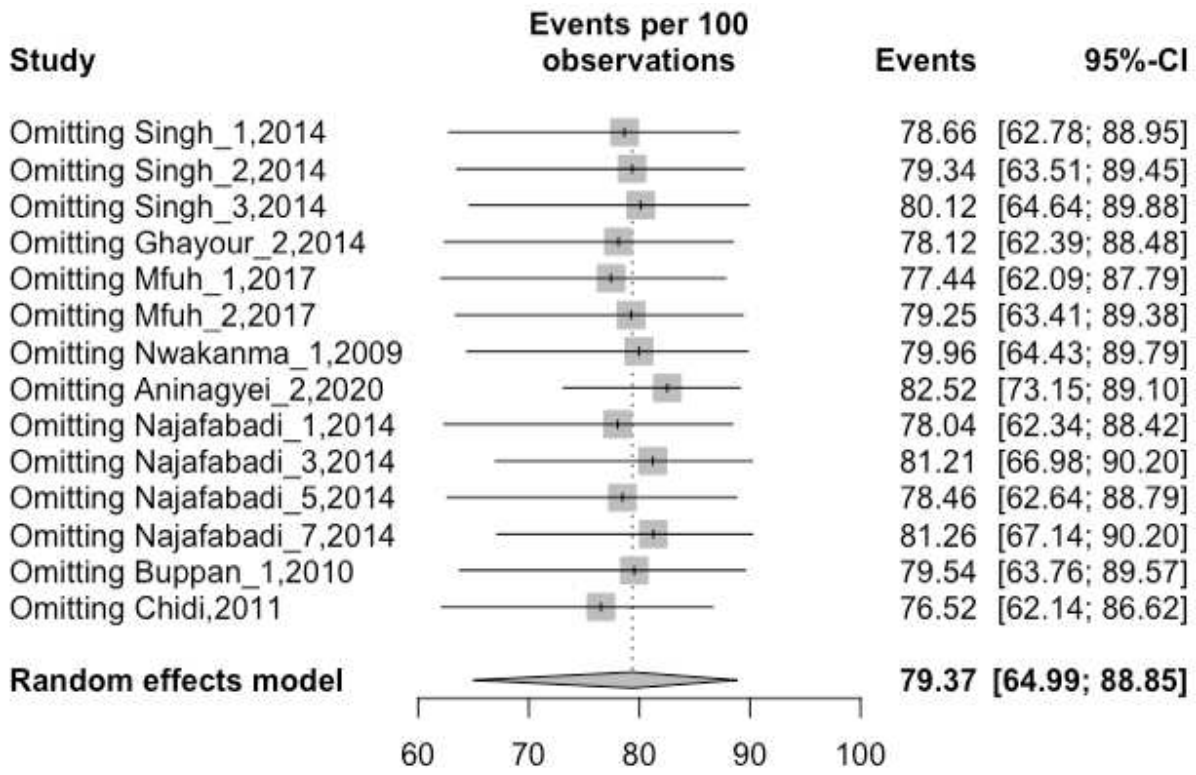
Supplementary Figure 8. Specificity of "sniff and tell" method



Supplementary Figure 9. DOR of "sniff and tell" method

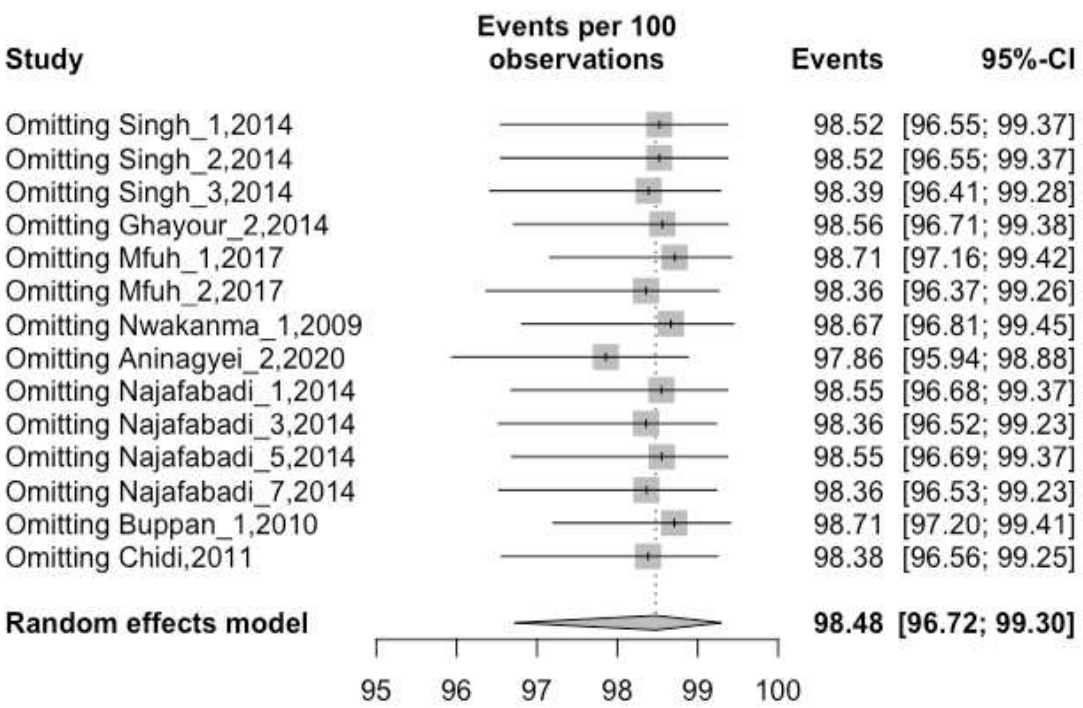
Leave-one-out analysis for studies conducted on saliva

Sensitivity



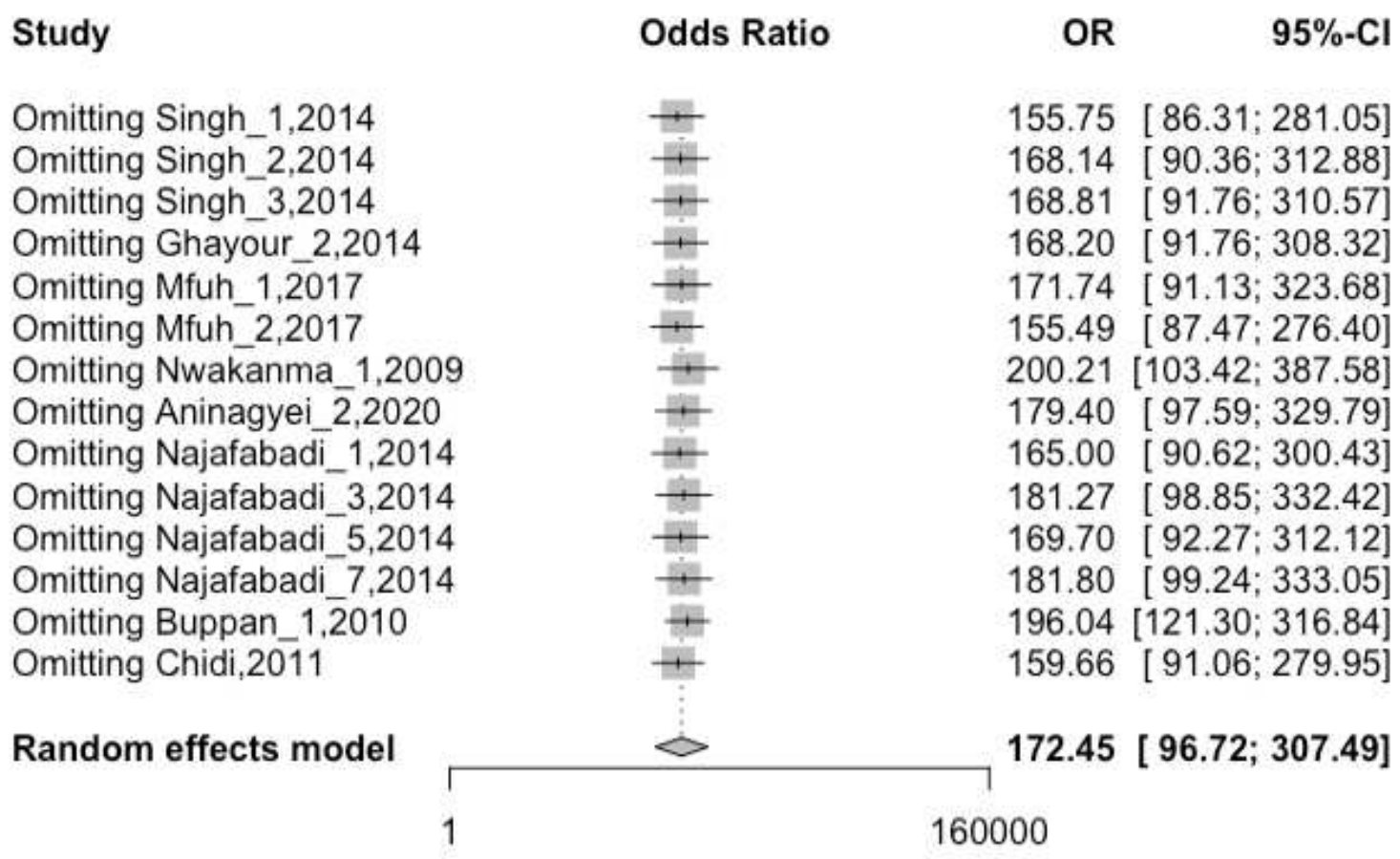
Supplementary Figure 10. Leave-one-out analysis of studies reporting on saliva (sensitivity)

Specificity



Supplementary Figure 11. Leave-one-out analysis of studies reporting on saliva (specificity)

DOR

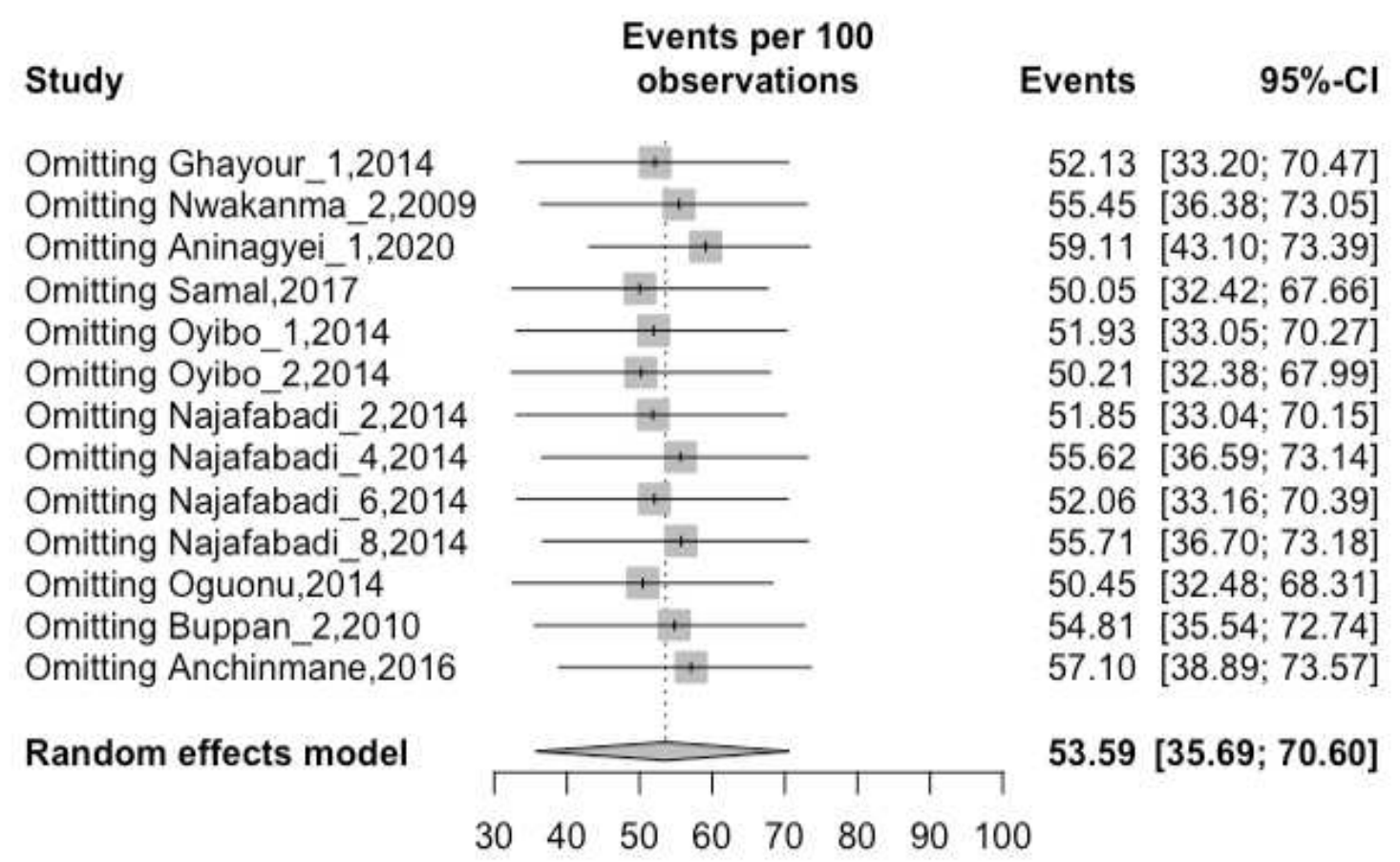


Supplementary Figure 12. Leave-one-out analysis of studies reporting on saliva (DOR)

NB: the leave-one-out analysis was conducted regardless of the reference test used in blood.

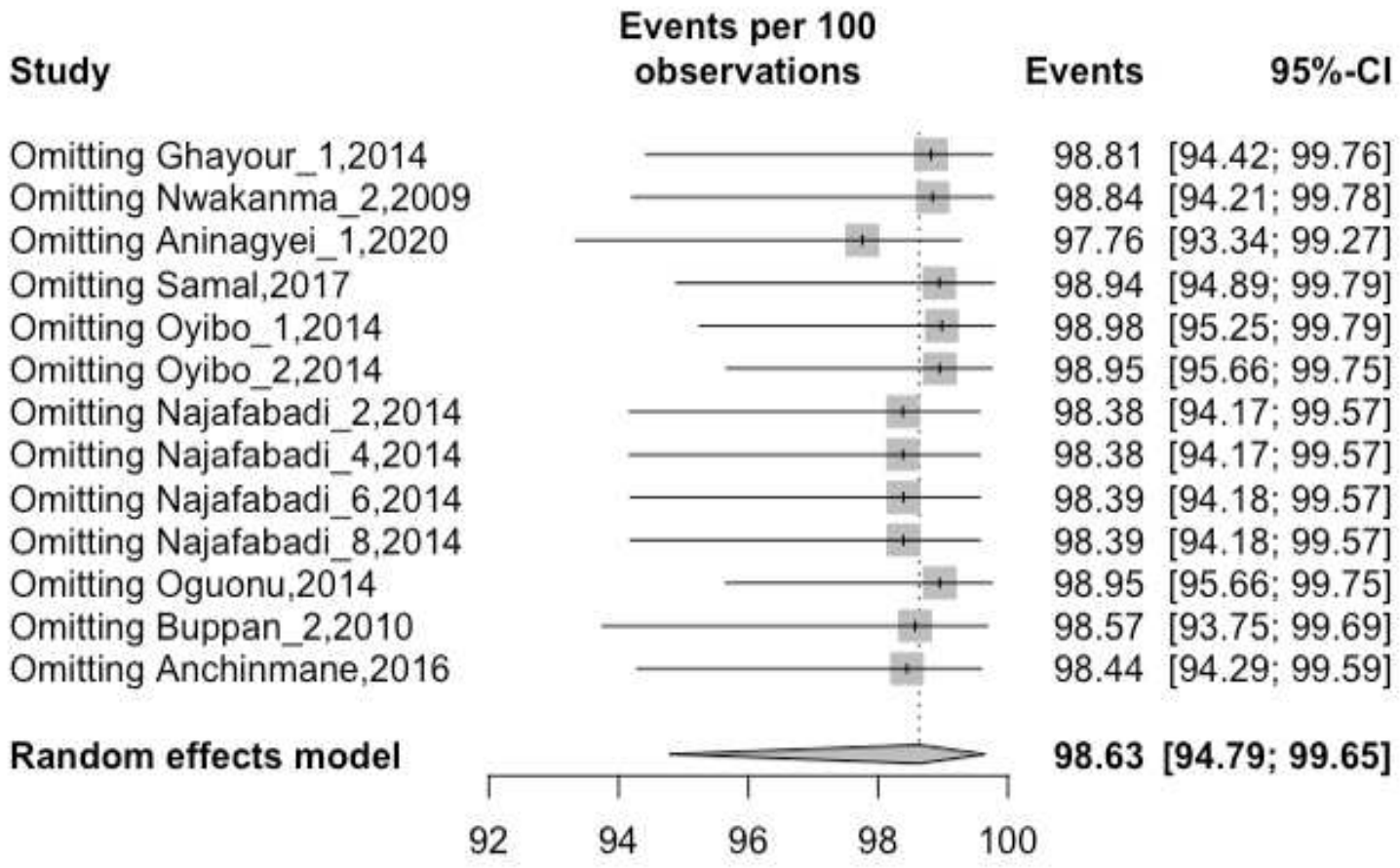
Leave-one-out analysis for studies conducted on urine

Sensitivity



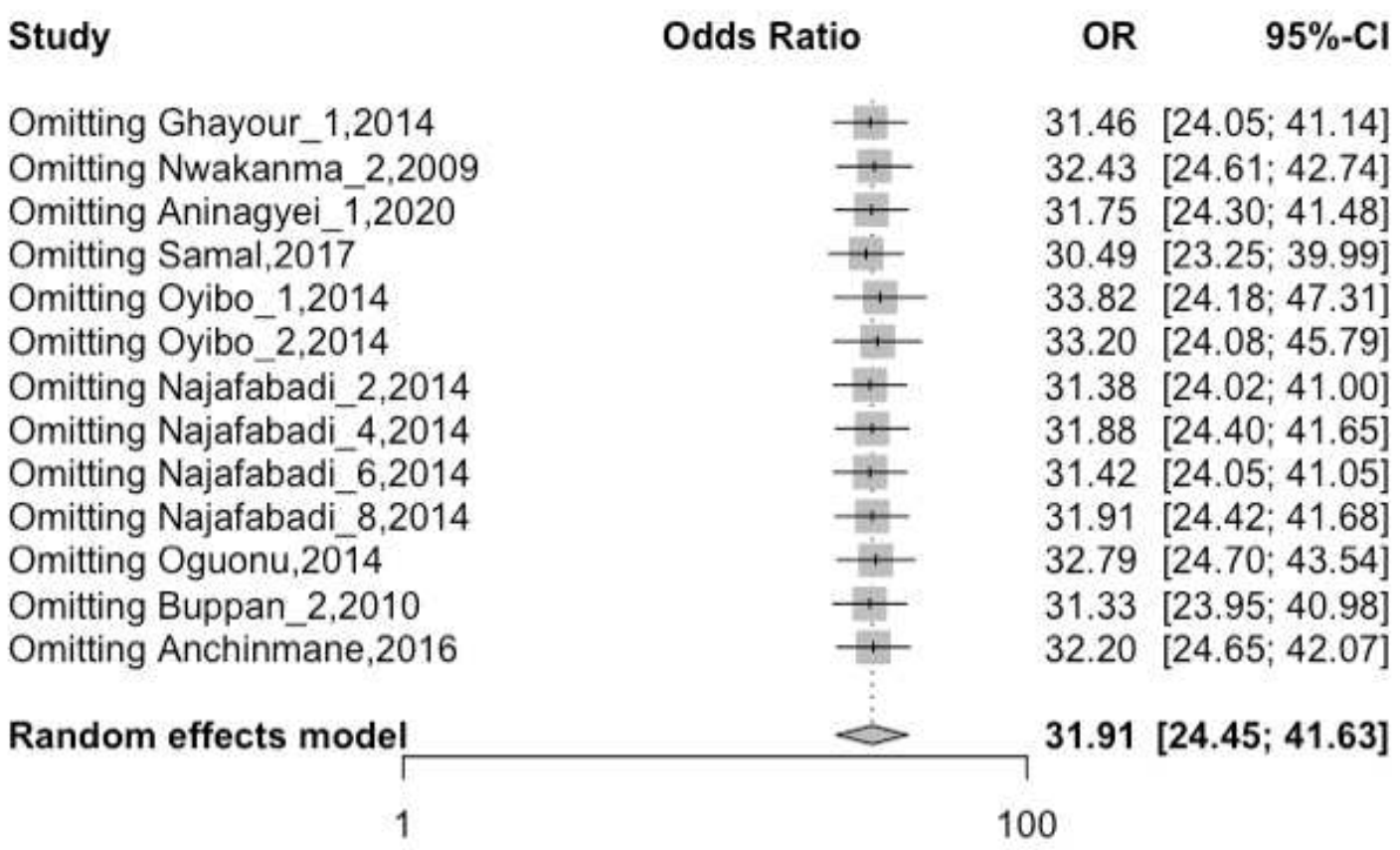
Supplementary Figure 13. Leave-one-out analysis of studies reporting on urine (sensitivity)

Specificity



Supplementary Figure 14. Leave-one-out analysis of studies reporting on urine (specificity)

DOR



Supplementary Figure 15. Leave-one-out analysis of studies reporting on urine (DOR)

NB: the leave-one-out analysis was conducted regardless of the reference test used in blood.

Danwang C, *et al. BMJ Global Health* 2021; 6:e005634. doi: 10.1136/bmjgh-2021-005634